

Exp-3: Generative Adversarial Network

AIM:

To implement a **Generative Adversarial Network (GAN)** to generate synthetic images using the MNIST dataset.

DATASET:

MNIST Handwritten Digits Dataset

PROCEDURE:

Step 1: Import Libraries

```
import torch

import torch.nn as nn

import torchvision

import torchvision.transforms as transforms

import matplotlib.pyplot as plt
```

Step 2: Load Dataset

```
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5,), (0.5,))])

dataset = torchvision.datasets.MNIST(root='./data', download=True, transform=transform)

dataloader = torch.utils.data.DataLoader(dataset, batch_size=64, shuffle=True)
```

Step 3: Define Generator

```
class Generator(nn.Module):
    def __init__(self):
        super(Generator, self).__init__()
```

```
self.model = nn.Sequential(nn.Linear(100, 256), nn.ReLU(), nn.Linear(256, 512), nn.ReLU(), nn.Linear(512, 784), nn.Tanh() )  
def forward(self, x):  
    return self.model(x)
```

Step 4: Define Discriminator

```
class Discriminator(nn.Module):  
    def init(self):  
        super(Discriminator, self).init()  
        self.model = nn.Sequential(nn.Linear(784, 512), nn.LeakyReLU(0.2), nn.Linear(512, 256), nn.LeakyReLU(0.2),  
nn.Linear(256, 1), nn.Sigmoid() )  
  
    def forward(self, x):  
  
        return self.model(x)
```

Step 5: Initialize Models

```
generator = Generator()  
discriminator = Discriminator()
```

Step 6: Loss and Optimizer

```
criterion = nn.BCELoss()  
g_optimizer = torch.optim.Adam(generator.parameters(), lr=0.0002)  
d_optimizer = torch.optim.Adam(discriminator.parameters(), lr=0.0002)
```

Step 7: Training GAN

```
epochs = 10  
  
for epoch in range(epochs):  
  
    for images, _ in dataloader:  
  
        images = images.view(-1, 784)  
        real_labels = torch.ones(images.size(0), 1)  
        fake_labels = torch.zeros(images.size(0), 1)  
        outputs = discriminator(images)
```

```

d_loss_real=criterion(outputs, real_labels)
noise = torch.randn(images.size(0), 100)

fake_images = generator(noise)

outputs = discriminator(fake_images.detach())

d_loss_fake = criterion(outputs, fake_labels)

d_loss = d_loss_real + d_loss_fake d_optimizer.zero_grad()
d_loss.backward()
d_optimizer.step()

outputs = discriminator(fake_images)
g_loss = criterion(outputs, real_labels)
g_optimizer.zero_grad()
g_loss.backward()
g_optimizer.step()

print(f"Epoch [{epoch+1}/{epochs}] | D Loss: {d_loss.item()} | G Loss: {g_loss.item()}")

```

Step 8: Generate Images

```

noise = torch.randn(16, 100)

generated_images = generator(noise).detach()
generated_images = generated_images.view(-1, 28, 28)
fig, axes= plt.subplots(4, 4, figsize=(5,5))
for i, ax in enumerate(axes.flat):
    ax.imshow(generated_images[i], cmap='gray')
    ax.axis('off')
plt.show()

```

OUTPUT:

```
*** Using: cpu
Epoch [1/5] | D Loss: 0.9090 | G Loss: 1.8620
Epoch [2/5] | D Loss: 1.0690 | G Loss: 2.4561
Epoch [3/5] | D Loss: 1.1777 | G Loss: 0.6676
Epoch [4/5] | D Loss: 1.0934 | G Loss: 0.8375
Epoch [5/5] | D Loss: 1.0666 | G Loss: 1.2042
```



RESULT:

The GAN successfully learns to generate handwritten digit images similar to MNIST. Initially, outputs are noisy, but improve with training as the generator and discriminator compete.